Software Requirements Specification (SRS)

Monthly AI-Powered Fashion Trend Recommendation Website

Field	Information
Team Name	hew khao mak
Project Title	Monthly AI-Powered Fashion Trend Recommendation Website
SDLC Chosen & Reason	Agile Model (Scrum Framework) — Chosen because the AI fashion trend recommendation system involves high technical complexity and must adapt rapidly to fast-changing fashion trends. Agile (Scrum) provides maximum flexibility for iterative feedback and continuous updates.
Date / Version	v1.0 · August 27, 2026
GitHub Repo	https://github.com/pun-code/hew-kaow-makk

1. Introduction

1.1 Purpose and Scope

Consumers and fashion enthusiasts struggle to keep pace with rapidly evolving monthly fashion trends, leading to hesitation and wasted time when selecting outfits or purchasing apparel. This project aims to address these challenges by building a web application that leverages a custom-trained AI model to automatically analyze, aggregate, and summarize the latest monthly fashion trends. The platform provides personalized, highly accurate, and up-to-date style recommendations tailored to individual user preferences.

1.2 Users and Stakeholders

Name / Role	Kind	What they need	What they fear / forbid
Fashion Shopper / General User	Primary user	Quickly discover personalized monthly fashion trends without searching across multiple platforms.	Receiving irrelevant recommendations, outdated trend data, or navigating an overcomplicated UI.
Fashion Creator / Brand Merchant	Secondary stakeholder	Opportunities to feature authentic brand items and reach target consumers searching for active trends.	Misrepresented product data or inaccurate trend direction generated by the AI system.
AI Data & Infrastructure Provider	Supplier	Stable compute execution, structured API interaction, and efficient data request management.	API quota exhaustion, unexpected compute cost spikes, or unauthorized usage of copyrighted fashion datasets.

2. Overall Description

2.1 Product Context — In Scope & Out of Scope

IN SCOPE (MVP Features)	OUT OF SCOPE (Explicit Exclusion)
• FR-1: AI body measurement & skin tone style analysis (5 style recommendations) and outfit preview image generation (with guest mode support) • FR-2: Body measurement-based search for existing styles on the website • FR-3: Personal outfit bookmarking to save favorite outfits • FR-4: User login (Email/Password and Google OAuth) to save bookmarked outfits • FR-5: Bilingual language translation system (TH / EN) for UI and AI-generated style descriptions	• In-app e-commerce transaction or payment gateway system • Style sharing system allowing users to upload and post their outfits • In-app user-to-user messaging or chat system

2.2 Assumptions and Constraints

> **Assumption:** Users access the application via modern desktop or mobile web browsers with an active internet connection.

> **Constraint (AI Dataset & Generation):** The AI style analysis and outfit preview image generation models must be trained and operated using legally compliant or publicly available fashion datasets and imagery.

> **Constraint (UI & AI Output Language):** The system interface and AI-generated style descriptions support English (EN) and Thai (TH) language localization.

> **Constraint (Data Privacy):** Only registered user login credentials (Email/Password or Google OAuth), body measurements, skin tone inputs, and bookmarked outfit IDs are stored. Guest trial sessions do not persist personal data, and no financial or payment data is collected.

> **Constraint (Timeline):** 10-week build window shared with four other milestones (M1–M5).

3. Functional Requirements

> ### FR-1 — AI Body Measurement & Skin Tone Analysis and Outfit Preview Generation
>
> **User Story:** As a general user (including guest users without logging in), I want to input my body measurements (height, weight, chest, waist, hip) and skin tone so that the AI recommends 5 suitable fashion styles for me to select 1 and generates an outfit preview image matching my body proportions and chosen style.
>
> **Pain Traces To:** Consumers struggle to determine which fashion styles suit their specific body proportions and skin tone, and cannot visualize how an outfit will look on them before choosing.
>
> **Acceptance Criteria:**

- Given a guest or logged-in user enters their height, weight, chest-waist-hip measurements, and skin tone, when submitted, then the AI analyzes the inputs and displays 5 suitable fashion style recommendations for the user to choose from.
- Given the 5 recommended styles are displayed, when the user selects 1 style, then the AI generates and renders an outfit preview image reflecting both the user's inputted body measurements/skin tone and the selected style.

FR-2 — Body Measurement-Based Style Search

User Story: As a general user, I want to input my body measurements to search existing styles available on the website, so that I can find outfits that fit my body proportions.

Pain Traces To: Hesitation and wasted time browsing general fashion catalogs that do not match an individual's body measurements.

Acceptance Criteria:

- Given a user enters their body measurements into the search form, when matching styles exist on the website, then the system displays the list of existing styles matching those measurements.
- Given a user searches with body measurements that have no exact match on the website, when results are displayed, then the system shows a clear "no matching styles found" message alongside similar alternative styles.

FR-3 — Personal Outfit Bookmark

User Story: As a logged-in user, I want to bookmark favorite outfits to my personal list, so that I can save and review outfits I like later.

Pain Traces To: Forgetting previously liked outfits or styles, forcing users to repeat searches.

Acceptance Criteria:

- Given a logged-in user clicks the bookmark button on an outfit, when confirmed, then the outfit appears in the user's personal bookmark list immediately.
- Given a logged-in user removes a bookmarked outfit, when clicked, then the outfit is removed from their personal bookmark list immediately.

FR-4 — User Authentication and Bookmark Persistence

User Story: As a user, I want to log in using Email/Password or Google authentication, so that my bookmarked outfits are securely saved to my personal account across sessions.

Pain Traces To: Losing bookmarked outfits when closing the browser or switching devices because no user account persistence exists.

Acceptance Criteria:

- Given a user accesses the login screen, when they authenticate using valid Email/Password credentials or Google Login, then the system signs them in and loads their saved bookmark list.
- Given a logged-in user saves outfits to their bookmarks and later logs back into their account, when the bookmark section loads, then all previously bookmarked outfits remain persisted.

FR-5 — Bilingual Language Translation (TH / EN)

User Story: As a Thai or international user, I want to switch the system language between Thai (TH) and English (EN) for both the website interface and AI-generated style descriptions, so that I can understand all recommendations comfortably.

Pain Traces To: Language barriers preventing Thai or international users from clearly understanding website navigation and AI-generated styling explanations.

Acceptance Criteria:

- Given a user selects Thai (TH) or English (EN) from the language switcher, when toggled, then all website menus, labels, and buttons update immediately to the selected language.
- Given an AI style analysis description is displayed, when the user switches the language between Thai (TH) and English (EN), then the AI-generated style explanation is displayed in the selected language.

4. Non-Functional Requirements

ID	Kind	Statement (Testable)	How We Will Check
NFR-1	Performance	The monthly fashion trend recommendation page shall load and render completely within 3.0 seconds.	Chrome DevTools Network/Performance Audit (5 trial average).
NFR-2	Accuracy / AI	The AI style recommendation and outfit image generation model shall achieve a style and body-measurement alignment accuracy of at least 80% on test dataset evaluations.	Model evaluation script run against a validation dataset of 100 body-measurement and style test cases.
NFR-3	Usability	A first-time user shall successfully search and bookmark a style within 45 seconds without consulting a manual.	Timed user trial with 3 first-time participants.
NFR-4	Availability	The website application shall maintain an operational availability uptime of at least 99.0% per calendar month.	Uptime monitoring service log review.
NFR-5	Security / Privacy	A logged-in user shall access only their own account profile and bookmarked outfits; direct requests modifying or accessing another user's ID shall be rejected.	API security test altering userId parameter, verifying 403 Forbidden response.

5. Use Cases

5.1 Use Case List

Use Case	Actor	Related FR	Brief Success Story
UC-1: Analyze Body & Generate Outfit Preview	Guest User / Registered User, AI Pipeline	FR-1	User inputs height, weight, chest-waist-hip measurements, and skin tone; AI suggests 5 styles, user selects 1, and AI generates an outfit preview image.
UC-2: Search Styles by Measurements	Guest User / Registered User	FR-2	User enters body measurements to search existing website styles and views exact matches or a not-found notice with similar styles.
UC-3: Bookmark Favorite Outfit	Registered User	FR-3	Logged-in user clicks bookmark on an outfit, adding it to their personal bookmark list.
UC-4: User Login & Manage Account	Registered User	FR-4	User logs in via Email/Password or Google OAuth to save and retrieve their personal bookmarked outfits.
UC-5: Switch Language (TH / EN)	Guest User / Registered User, AI Pipeline	FR-5	User switches language between TH and EN, updating both website UI text and AI-generated style descriptions.

5.2 Use-Case Diagram

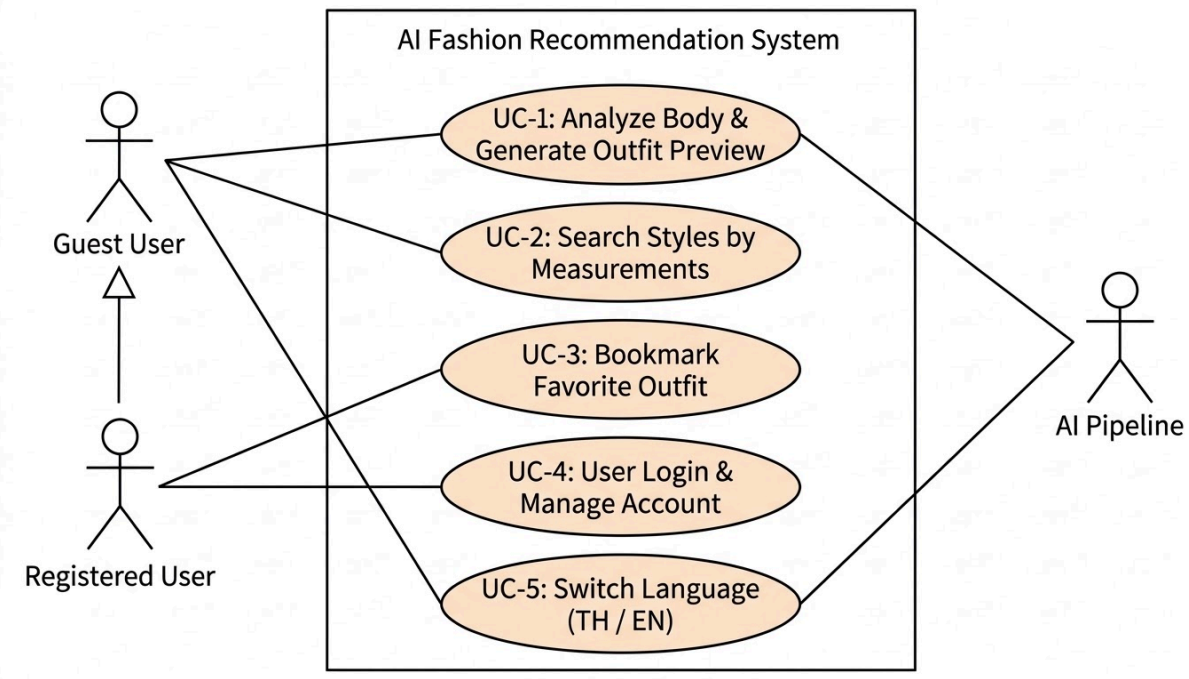


Figure 1: Use Case Diagram mapping Guest User, Registered User, and AI Pipeline across UC-1 to UC-5.

6. Out of Scope

- In-app e-commerce transactions or payment gateway integration.
- Style sharing system allowing users to upload and post their outfits.
- In-app user-to-user messaging or social chat features.

7. Traceability Matrix (Golden Thread)

Problem / Pain	Requirement (FR)	Solution / Design	Feature Built	Test
Users cannot visualize how styles fit their body measurements and skin tone	FR-1 AI Body & Skin Tone Analysis + Outfit Preview	BodyProfile Entity + AI Style & Image Generation Pipeline	Guest Body Analysis & 5-Style Preview Generator	Unit Test: analyze_body_and_generate_preview()
Browsing general catalogs wastes time when styles do not fit user measurements	FR-2 Measurement-Based Style Search	(Week 5)	(Week 6–7)	(Week 7)
Forgetting previously liked outfits or styles	FR-3 Personal Outfit Bookmark	(Week 5)	(Week 6–7)	(Week 7)
Losing saved bookmarked outfits across browser sessions or devices	FR-4 User Login & Bookmark Persistence	(Week 5)	(Week 6–7)	(Week 7)
Language barriers reading website UI and AI style explanations (TH / EN)	FR-5 Bilingual Translation (TH / EN)	(Week 5)	(Week 6–7)	(Week 7)

8. AI Usage Log

Date	Tool / Model	What We Asked	What AI Produced	What We Changed / Verified	What We Learned
27 Aug 2026	Gemini	Draft problem statements, refine project scope, and structure SRS for AI fashion recommendation website.	Drafted SRS structure, User Stories, NFRs, and Use Cases. Generate use case diagram,	Refined custom AI model constraints, changed purchase model to official store redirection, and adjusted UC-4 wording.	AI speeds up document structuring; scope and technical details must be human-verified.

9. Self-Check

☑ Every FR is a testable User Story with clear benefit ("so that...") tied to a stated pain.

☑ Every FR includes at least two measurable Acceptance Criteria.

☑ No unquantified words ("fast", "easy") are used in requirements.

☑ NFRs are defined, including mandatory security/privacy access controls (NFR-5).

☑ Every FR traces back to an identified problem in the Traceability Matrix.

☑ Scope and Out-of-Scope lists are explicitly documented.

☑ Use Case list directly maps to the functional requirements.

☑ AI Usage Log is accurately and transparently maintained.